



AI CLINIC ASSISTANT

AI AGENT



SEPTEMBER 29, 2025

APP AURA

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Introduction:

The AI Clinic Assistant is a smart tool that helps clinics manage patients more easily. Clinics often struggle with long wait, missed appointments and busy staff. Our AI appointment assistant solves this by handling appointments, reminders, patient queries (Complaints, Updates, Cancellation and if they require any kind of help they are struggling with). This saves time, reduces workload, reduces the cost of too much staff and gives the patient a faster and more affordable experience.

Problem Statement:

Many hospitals face the issue of appointment handling sometimes it's missed, sometimes when a receptionist is busy in a call and at that right moment some other calls they can't book an appointment, also hospitals have 3-4 receptionists minimum per shift of 8 hours increasing the cost of the hospitals. (To check the cost details proceed at end).

Solution:

For these kind of problems we have our Ai agent which can handle these things easily, fully integrated with security. This service is for user's, for admin and for receptionist's. This service will lower the cost of staff from 40-50% keeping hospital's working more easy and better for user-experience.

(Features are below this page).

Tech Stack:

1. Python
2. FastApi
3. MongoDB
4. React js
5. Vapi
6. Make.com

Features:

This Assistant have 3 parts we will see 1 by 1.

For Admin:

1. Check history (Add update cancel) of Appointments.
2. Check history (Add update cancel) of chats and call logs (by decrypting)
3. Check cost on his dashboard that Ai is utilizing.
4. Check how many users are using this service.
5. Can also updates the bot configuration's like voice and prompt if he wants to add something.
6. Only Admin can see the history of how users are utilizing AI, how they chat, how they talk, all data is encrypted and only admin can decrypt it by logging in.

For User:

1. Can book an appointment by chat or by call.

2. Can update or cancel an appointment by chat or call.
3. Can see the history of chats and calls on his dashboard.
4. Will receive a confirmation email.
5. Can see his appointment history.

For receptionist:

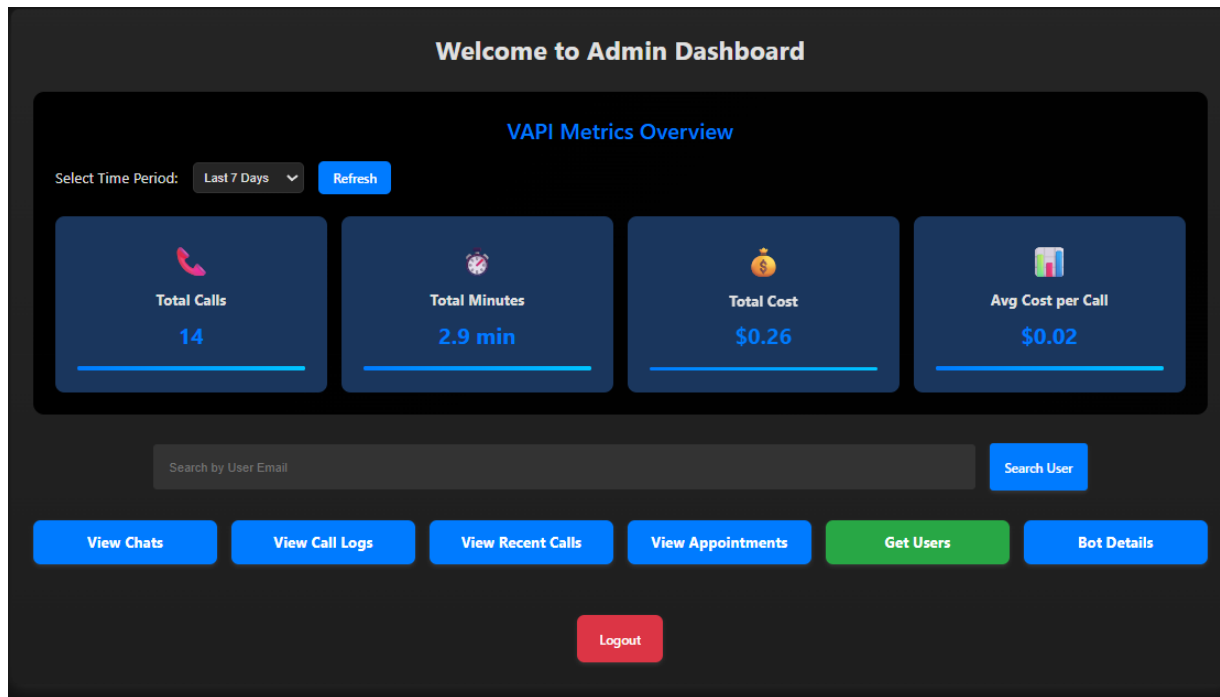
1. Can see only the Appointment details, cancellations and updates.

Ai Agent:

1. Inbound calls.
2. Automatic outbound calls.
3. Chat
4. Sending Automatic confirmation emails.
5. Handle every type of user request.

Flow of Service

Starting with Admin

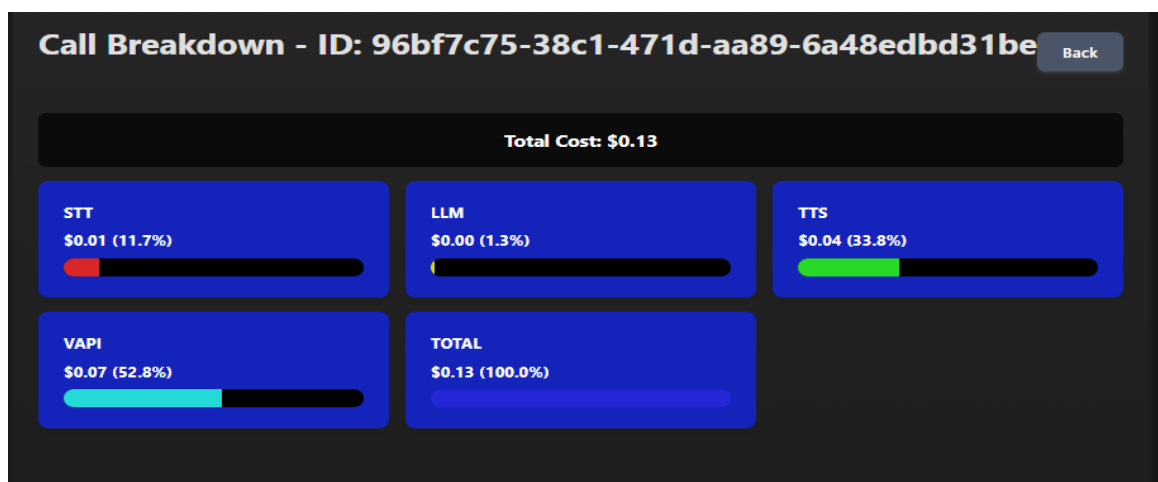


The calls, minutes cost you seeing is the cost of Ai and total calls done to Ai agent.

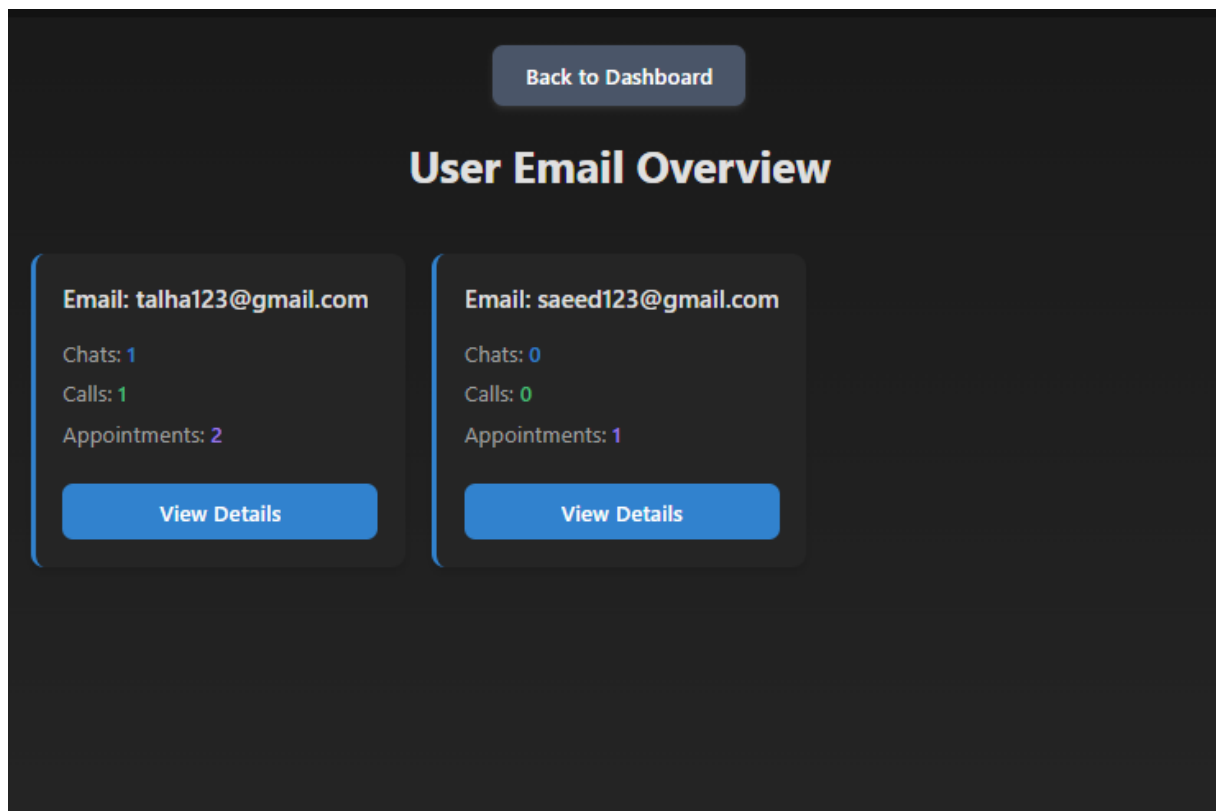
View chat → you can see all users chat.

View call logs → you can see call logs transcripts etc.

View Recent calls → You can see the last 10 or selected calls number and breakdown its call cost like this.



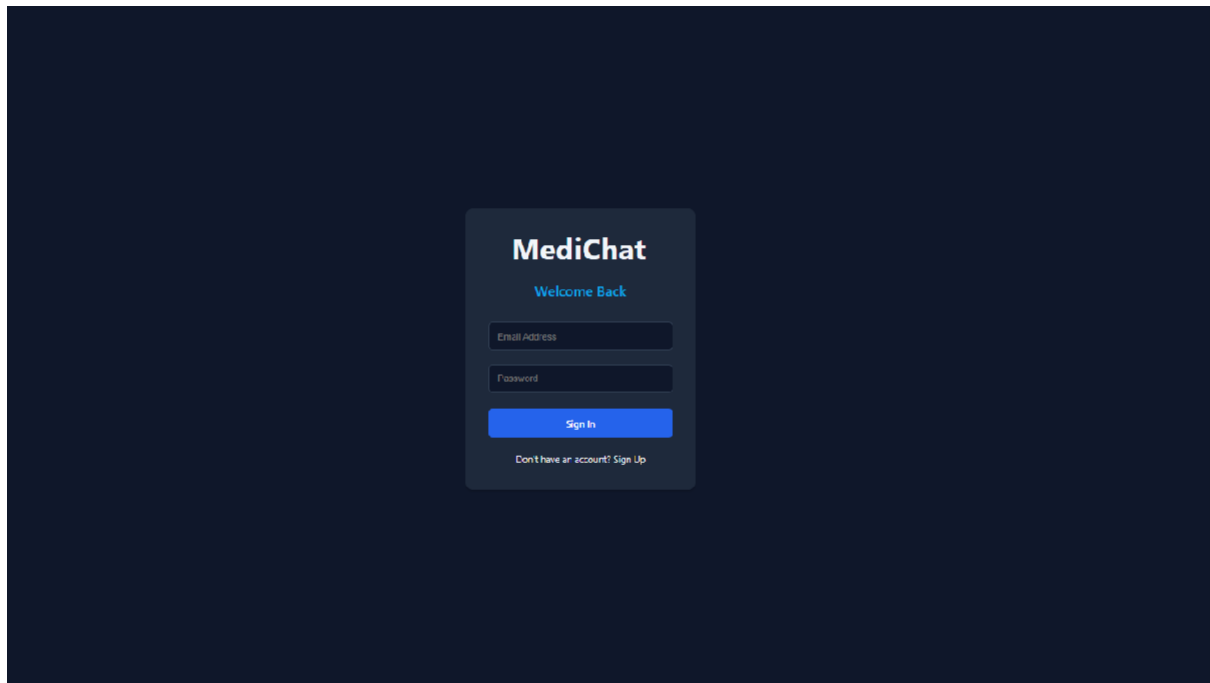
Get User → you can see all users with details like:



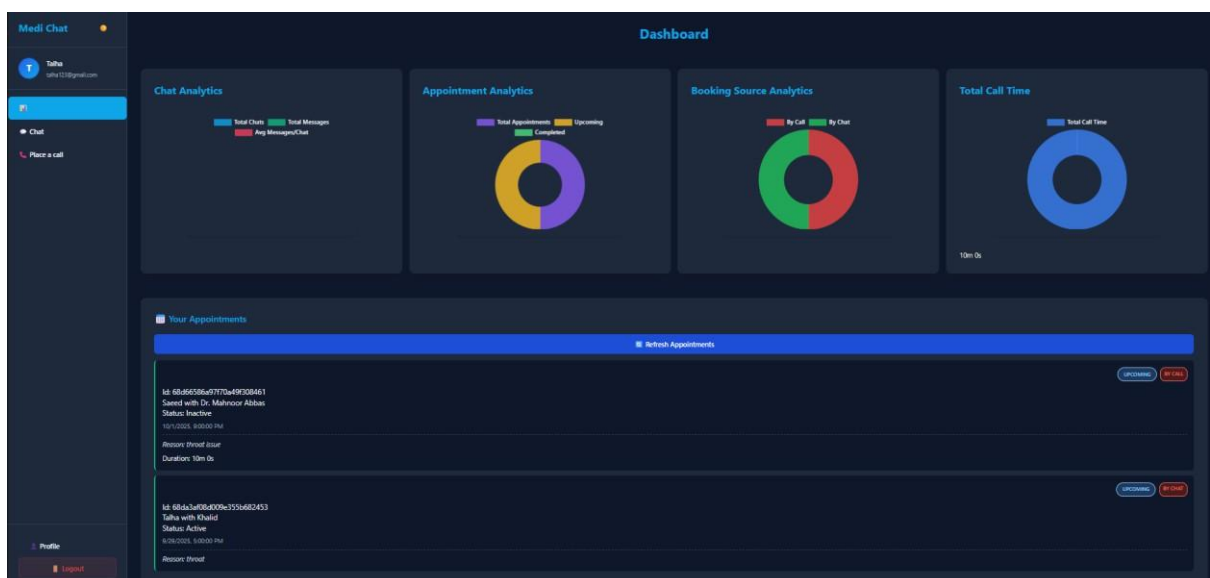
Bot Details → you can update the bots configurations like voice, transcriber with cost detection to lower the cost and latency.

User's side:

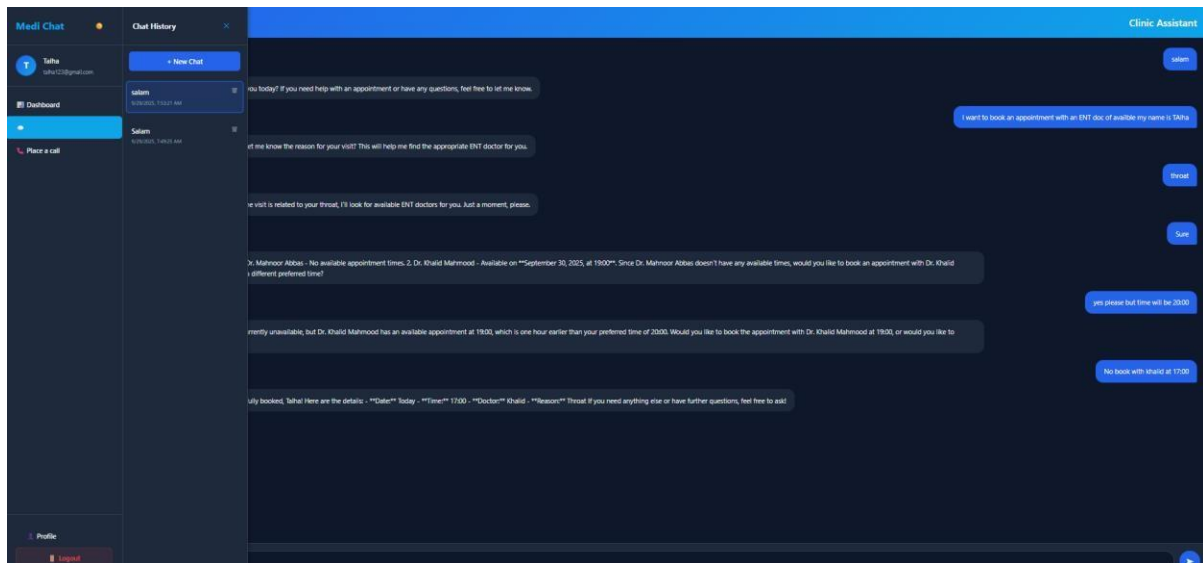
User login page →



User dashboard → you can see appointments upcoming, recent, Active inactive.



Chat button → you can chat with AI to book also history are saved for data security.



Call button → you can call on website with AI no phone number needed and also all the transcript is saved in database.

Chat Scenario:

User wants to book an appointment with an ENT doctor. Having throat issue. Vapi have the same data stored in database.

 Assistant

Thank you, Talha! Could you please let me know the reason for your visit? This will help me find the appropriate ENT doctor for you.

 User

throat

 Assistant

Got it, Talha! Since your reason for the visit is related to your throat, I'll look for available ENT doctors for you. Just a moment, please.

 User

Sure

 Assistant

[Tool Call] get_available_time with params {"doctor_name": "Dr. Mahnoor Abbas"}

 User

yes please but time will be 20:00

 Assistant

[Tool Call] get_available_time with params {"doctor_name": "Dr. Mahnoor Abbas"}

 User

No book with khalid at 17:00

salam

Wa alaikum salam! How can I assist you today? If you need help with an appointment or have any questions, feel free to let me know.

I want to book an appointment with an ENT doc of available my name is Talha

Thank you, Talha! Could you please let me know the reason for your visit? This will help me find the appropriate ENT doctor for you.

throat

Got it, Talha! Since your reason for the visit is related to your throat, I'll look for available ENT doctors for you. Just a moment, please.

Sure

I found two ENT doctors for you: 1. Dr. Mahnoor Abbas - No available appointment times. 2. Dr. Khalid Mahmood - Available on **September 30, 2025, at 19:00**. Since Dr. Mahnoor Abbas doesn't have any available times, would you like to book an appointment with Dr. Khalid Mahmood at 19:00, or do you have a different preferred time?

yes please but time will be 20:00

It looks like Dr. Mahnoor Abbas is currently unavailable, but Dr. Khalid Mahmood has an available appointment at 19:00, which is one hour earlier than your preferred time of 20:00. Would you like to book the appointment with Dr. Khalid Mahmood at 19:00, or would you like to choose a different time?

No book with khalid at 17:00

Your appointment has been successfully booked, Talha! Here are the details: - **Date:** Today - **Time:** 17:00 - **Doctor:** Khalid - **Reason:** Throat If you need anything else or have further questions, feel free to ask!

Here is the details in database and on user dashboard.

```

_id: ObjectId('68da3af08d009e355b682453')
patient_name : "Talha"
patient_email : "talha123@gmail.com"
doctor_name : "Khalid"
appointment_time : 2025-09-29T17:00:00.000+00:00
reason : "throat"
source : "book_chat"
call_id : "2b79583e-2afb-4749-9b3e-dfa761f1da09"
call_duration_minutes : 0
status : "Active"
call_duration_seconds : 0

```

Your Appointments

 Refresh Appointments

UPCOMING

BY CHAT

Id: 68da3af08d009e355b682453

Talha with Khalid

Status: Active

9/29/2025, 5:00:00 PM

Reason: throat

Now if he wants to cancel, just use that id showing on user dashboard. Again Vapi AI have same data stored.

 User

I want to cancel my appointment

 Assistant

I can help you with that! Could you please provide me with your appointment ID for verification?

 User

Sure here is my id

 Assistant

It seems like you didn't provide the appointment ID. Could you please share it so I can assist you with the cancellation?

 User

68da3af08d009e355b682453

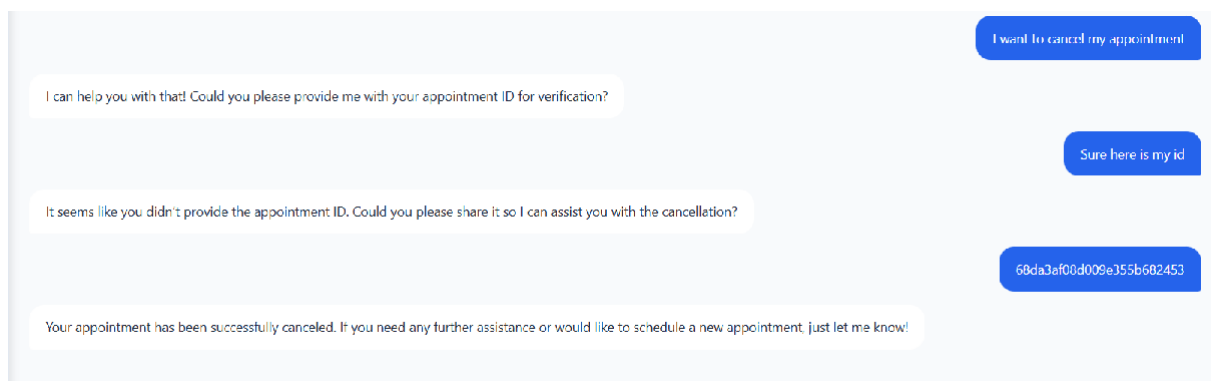
Output Messages 3

 Assistant

 Tool Calls

`</> delete_appointment`

```
{
  "id": "68da3af08d009e355b682453"
}
```



Now you can see status is Inactive now.

Now if user want to update the time of his appointment use that same id. Vapi have same messages stored.

 Assistant

Sure! To help you update your appointment, could you please provide me with your appointment ID and let me know what changes you would like to make?

 User

68da3af08d009e355b682453 just need to update the date to 2025-2025-10-01T1:00:00.000+00:00

 Assistant

[Tool Call] update_appointment with params {
 "id": "68da3af08d009e355b682453",
 "appointment_time": "2025-2025-10-01T1:00:00.000+00:00"
}

 User

2025-10-01 time will be 6pm

Output Messages 4

 Assistant

 Tool Calls

</> update_appointment

```
{  
  "id": "68da3af08d009e355b682453",  
  "appointment_time": "2025-10-01T18:00:00.000+00:00"  
}
```

```
_id: ObjectId('68da3af08d009e355b682453')
patient_name : "Talha"
patient_email : "talha123@gmail.com"
doctor_name : "Khalid"
appointment_time : 2025-10-01T18:00:00.000+00:00
reason : "throat"
source : "book_chat"
call_id : "2b79583e-2afb-4749-9b3e-dfa761f1da09"
call_duration_minutes : 0
status : "Active"
call_duration_seconds : 0
```

68da3af08d009e355b682453 just need to update the date to 2025-10-01T18:00:00.000+00:00

It looks like there was an issue with the date format you provided. Could you please confirm the new date you would like to set for your appointment? It should be in the format "YYYY-MM-DD".

2025-10-01 time will be 6pm

Your appointment has been successfully updated! Here are the details: - **Doctor:** Khalid - **New Date:** October 1, 2025 - **New Time:** 6:00 PM - **Reason for Visit:** Throat If you need any further assistance, feel free to ask!

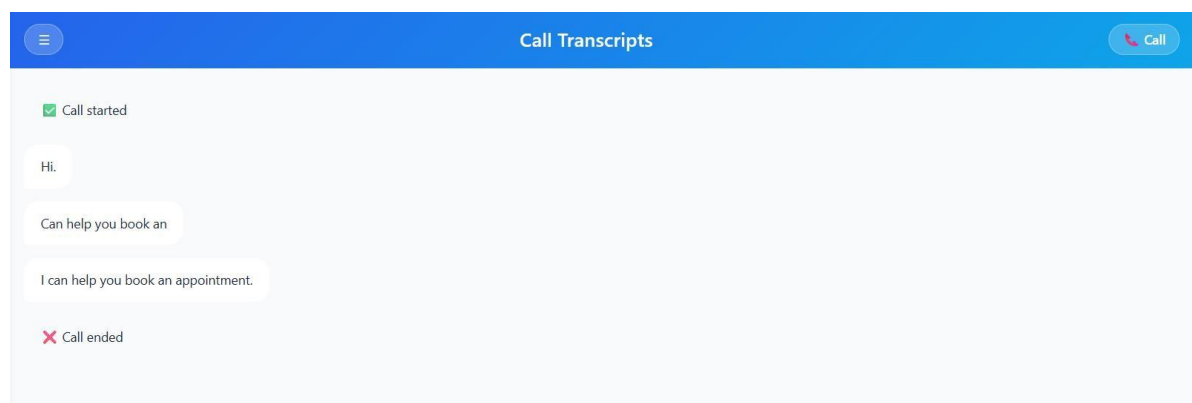
Messages of users are encrypted and only user can decrypt it when he logs in

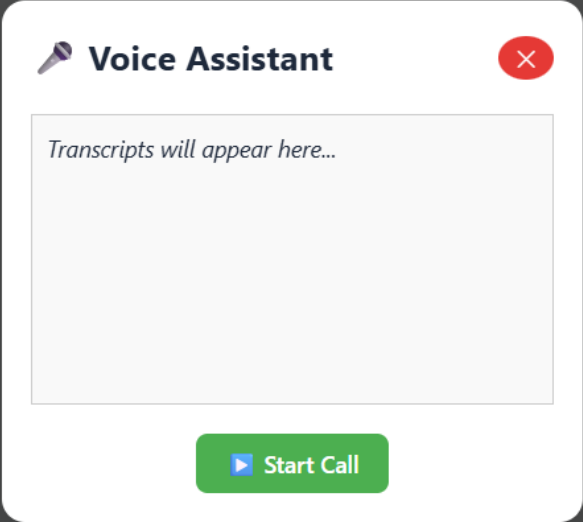
```

_id: ObjectId('68da3a108d009e355b68244f')
history_id : "4129ae2b-d78b-46bd-bd36-1cf259d04ec1"
chat_id : "fc036438-57ae-4ed4-8064-2ceadfe1705d"
title : "salam"
user_id : "68da39d98d009e355b68244c"
email : "talha123@gmail.com"
name : "talha123"
▼ messages : Array (12)
  0: "gAAAAABo2joQ2j-SuRErvZx2fkgIt88ytEZSsIIE02k_d-f0zheb69c9YpDx2valb-WHcW..."
  1: "gAAAAABo2joQfr7IhM89w-cXADF-iKlp7WHqxfv1A04Aq11b4gSmZd37NhVhWrF1rHqtIX..."
  2: "gAAAAABo2jqUnIW6wHLlbBMZjuFjbJ1-1fCbBYDYNRjvRTa3AMUW2ZNmzlagtzDmS3sIni..."
  3: "gAAAAABo2jqUuvoAphqg0u5t7BMDId3j5hxrAFPbC2k8i1QZNI_8refquh-1FYnVEIM-3K..."
  4: "gAAAAABo2jqsk1GxEH3Wr15Aat7fBcjd2Yxzw2GnGGNPa9MRR5ZqR0WrWxejRxSRcBbpEB..."
  5: "gAAAAABo2jqs-FEZr0rrs4urdmFj3_tyy1lfjSRI0pNLRXKaJz0vYYdmXuxbOCMDQYBJPP..."
  6: "gAAAAABo2jq5sq1PFpi5U1sd6_d7BUjGE9Qfa1IFwAkFQUxm7xpNH2Co6MnXMFUPCZ9rV4..."
  7: "gAAAAABo2jq50bawYSLJy9sYAYjnSqRJABm9uhdZc4UjLNX0qwtmNndswMGYkQXP20-rxL..."
  8: "gAAAAABo2jrc9XsgpeJRBnTrJxuDFkTpPEcvNY4Fteview03foY_vDaokmN5G40kGeKj0K9..."
  9: "gAAAAABo2jrcFs_qz7Qcz2889N239JY5TCVmgJtwGy1-HIAw4-QJHf8NIIahHT3gvbdX9n..."
  10: "gAAAAABo2jrxtpsVE9Dn8J6GJaGpUUF_dwBvUbdU33QEuDGKBFaNV984KyVTVIninMS5Rk..."
  11: "gAAAAABo2jrx4DWkG6P_ovSffa69lZIA1kku19QZuOXdVaRnYiRZKmxIWMUrdAj-bPzVSh..."
timestamp : "2025-09-29T07:53:21.050350"
```

Call Scenario 1:

Go on user dashboard and you can see the calls button like that at top right corner just click it and you will get a pop-up like:





Here is call logs database everything is encrypted same like chat only user with that email can decrypt it to see the transcript.

```

  _id: ObjectId('68da6c4184917025a7d6e2b2')
  body: Object
    message: Object
      timestamp: 1759145024807
      type: "end-of-call-report"
    analysis: Object
    artifact: Object
      startedAt: "2025-09-29T11:22:18.758Z"
      endedAt: "2025-09-29T11:23:40.351Z"
      endedReason: "customer-ended-call"
      cost: "gAAAAABo2mxBkw8Fc37fnWndIeHaEnLJkumWcRZF2Mhr8_0-cSg0lnlTuGjRk6jK9t0iw..."
      costBreakdown: "gAAAAABo2mxBB9sCAYhMIZbSa-6uH85YiAJIZUTKjdSgCv_e4vciETdbZS0g_FbZ2m1rFE..."
      costs: "gAAAAABo2mx87VV_F3nmGIWK-EtcxKV6jiXRBTgGCWeKVzxXKuCj8vc7JN_3-t7z5-dZSQ..."
      durationMs: 81593
      durationSeconds: 81.593
      durationMinutes: 1.3599
      summary: "gAAAAABo2mxBwEY5VKEhDhVwbpGGEJFGgmhwJ5U15F6MPFQxGmgr985vdr4dW0n1mkzu1p..."
      transcript: "gAAAAABo2mxBTZAMUNPoqjU9adiGKesP38Q4c8U8WKLJvf56YZk56KJm83AFTgfbQhavam..."
    messages: Array (18)
      0: Object
      1: Object
      2: Object
      3: Object
      4: Object
      5: Object
      6: Object
      7: Object
      8: Object
      9: Object
      10: Object

```

fced8840-1790-4fa8-a8d4-040b8b8f24c3-1759145022866-d34f19c8-1487-477a-905f-9947780efff2-stereo.wav

Call Scenario 2:

User calls with his phone number → To Ai Agent number → Same things happened → Same database scenario → both calls are same difference is the user's reliability.

The screenshot displays a call recording interface with a waveform at the top and a transcript below. The transcript shows a conversation between an Assistant and a User. The Assistant asks for the user's name, the user provides 'John', and the Assistant asks for the reason for the visit. The user states they want to book an appointment. The Assistant then asks for the reason for the appointment. The user provides 'Great. We have 2 ENT specialists available. 1'. The Assistant then asks for the user's preference for which doctor they'd like to see and what time they'd like to schedule. The user provides 'Just booked with first 1. And the time should be first of October 2025. And time will be 9 PM.' The Assistant then confirms the booking.

Recording 01:22

Audio

Logs Transcripts Analysis Structured Outputs Messages Call Cost

Assistant
Hi. I can help you book an appointment. Who am I speaking with?

4:22:14 PM (+00:00:00)

User
My name is John.

4:22:19 PM (+00:00:75)

Assistant
Um, thank you, John. What brings you in today? Could you please share the reason for your visit?

4:22:22 PM (+00:03:25)

User
Yes. I want to book an appointment.

4:22:28 PM (+00:09:44)

Assistant
Thank you for letting me know, John. Could you please specify the reason for your appointment? This will help me ensure you see the right specialist.

4:22:55 PM (+00:36:45)

Get Doctors By Specialty Completed successfully 04:22 PM
4:22:57 PM (+00:38:51)

Assistant
doctor Mahanur Abbas. 2. doctor Khaled Mahmoud. Do you have a preference for which doctor you'd like to see? And what time would you like to schedule your appointment?

4:22:59 PM (+00:41:22)

User
Just booked with first 1. And the time should be first of October 2025. And time will be 9 PM.

4:23:10 PM (+00:51:99)

Assistant
Your appointment has been successfully booked with doctor Manur Abbas for October

How the data is encrypting and decrypting?

User chat or call → Send's to database and encryption key generates a cipher text that is unreadable → save in database.

User wants to see his history → Click on chat or call history → request goes to database he use that same encryption key to decrypt it → User can now his history by decrypting.

Email Scenario:

When the Make.com webhook received the data from database the same data of appointment. He will automatically sends the email to patient email he received in data. Like below:

The image shows a Make.com workflow and its resulting email. The workflow consists of two steps: 'Webhooks' (Custom webhook) and 'Email' (Send an Email). The 'Email' step is configured to send an email to the patient's email address received in the data. The email content is as follows:

Summary
✓ 1 operation | 1 credit used

Initialization
✓ Operation 1

INPUT

- Bundle 1: (Collection)
 - Copy recipient: (Array)
 - To: (Array)
 - Blind copy recipient: (Array)
 - From: empty
 - Content: Hi ,
 - saeed,
 - Your appointment with Dr. Dr. Mahnoor Abbas is confirmed at exact 2025-10-03T18:00:00
 - If you have any questions, please reply to this email.
 - Thanks!
 - Sender: empty
 - Headers: (Array)
 - Reply-To: empty
 - Subject: Appointment booking success for saeed
 - Priority: normal
 - In-Reply-To: empty
 - References: (Array)
 - Attachments: (Array)
 - Content Type: text

OUTPUT

- Bundle 1: (Collection)
 - Message ID: <6c9aecc8-3646-f750-efaf-144046047-7a@business.adh...

11:37 

  74 



Appointment booking success for saeed

Inbox



su92-bssem-f22-146 11:33 AM

to me 



Hi ,

saeed,

Your appointment with Dr. Dr. Mahnoor Abbas is confirmed at exact 2025-10-03T18:00:00

If you have any questions, please reply to this email.

Thanks!

 Reply

 Forward



Cost (Rough projection):

AI Cost	Hospital's receptionist cost	What hospitals are saving?
0.1\$/min	8 hours 3 shifts. 18\$/hour	Now hospital don't need 3 receptionist per shift. Now only needs 1 to see bookings and managing.
Let's say 1000 min AI utilized. 0.1x1000 =100\$	For 1 day with 2 receptionist per shift 36x24=864\$	For 1 day 18x24 = 432\$
Monthly let say 0.1X 20000=2000\$	Monthly it is 25920\$. Remember its according to 24 hours.	Monthly it is 12960\$. Remember its according to 24 hours.
For 1 year: 2000x12 = 24000\$	For 1 year 12960x12 = 311040\$	For 1 year 12960x12 = 155520\$

Now with 1 receptionist/shift no need of 2 now 155520+24000
=179,520\$

Before when two receptionist 311040 – 179520 = 131520\$

Usually hospitals have more than 2 receptionists per shift but we will considering minimum to show how much this will save you the cost and much better and easy 24/7 experience.

Conclusion:

Now with our Ai agent, With 3 receptionists at a time hospitals still missed some booking leading to cause an issue, long waits. Now with this service user just go to web/app to just chat and book an appointment and they can place a call too. With this service hospital's cost will be saved a lot and it's still better have 24/7 availability. So the main part of this service are:

For hospital → lowers the cost, easy handling of appointment.

For users → with internet can place call, chat and see his history